

WARDA SHAHZAD

PhD Candidate, Chemistry | Kesterite (CZTS), Thin-Film, Photovoltaics, Semiconductor

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RESEARCH PROFILE

PhD candidate in Chemistry at COMSATS University Islamabad, Abbottabad Campus, working on solution-processed $\text{Cu}_2\text{ZnSnS}_4$ (CZTS) kesterite thin-film solar absorbers. My doctoral research applies systematic alkali-cation doping (Li, Na, K) via aqueous chemical bath deposition to suppress Cu/Zn antisite defects and secondary phases (ZnS , Cu_2SnS_3), the primary loss mechanisms limiting CZTS device efficiency, and to tune the resulting structural, optical, and electrical properties. My characterization toolkit spans XRD (Williamson-Hall analysis, dislocation density, microstrain), SEM/EDX, UV-Vis/Tauc-plot modeling, dielectric and conductivity analysis, and DFT-based structural/DOS/band-structure interpretation. My thesis has been submitted for evaluation, and I am currently awaiting the final viva voce. I am seeking postdoctoral opportunities to extend this work to new synthesis platforms and semiconductor material systems.

EDUCATION

PhD, Chemistry (Materials Chemistry)

2020 – Present

COMSATS University Islamabad, Abbottabad Campus, Pakistan

Thesis: "Exploration of the Physical Properties of Kesterite Thin Films Modified by Alkali Metal Cations for Sunlight Harvesting." Status: submitted for evaluation; final viva voce pending.

MS, Chemistry

2017 – 2019

COMSATS University Islamabad, Abbottabad Campus, Pakistan

Thesis: "Visible Light Driven Photocatalysis Based on Alkaline Earth Metal Zirconate Perovskite/ Graphene Oxide Composite for Wastewater Treatment"

MSc, Chemistry

2015 – 2017

University of Wah, Pakistan

RESEARCH EXPERIENCE

PhD Researcher, Chemistry

2020 – Present

COMSATS University Islamabad, Abbottabad Campus

- Investigated Li, Na, and K alkali-cation doping in CZTS kesterite thin films for defect engineering and secondary-phase suppression, using a systematic three-metal comparison on the same aqueous chemical-bath-deposition platform.
- Developed a two-step thermal annealing protocol for Li-doped CZTS that separated dopant effects from thermal effects, reducing dislocation density from 7.3 to $2.13 \times 10^{14} \text{ m}^{-2}$ and tuning bandgap from 1.36 to 1.27 eV ; measured electrical conductivity of $3.6 \Omega^{-1}\text{cm}^{-1}$ for the optimized film.
- Showed Na doping narrows the optical bandgap from 1.5 to 1.0 eV while reducing microstrain from 0.37 to 0.13×10^{-3} , linking structural relaxation directly to optoelectronic performance.
- Demonstrated K doping increases crystallite growth from 40 to 52 nm alongside a bandgap shift from 1.50 to 1.42 eV , while retaining the tetragonal kesterite structure.
- Extended the doping study into Na/Li and Li/K co-doping systems, currently being prepared as an additional manuscript.
- Applied Williamson-Hall analysis, SEM/EDX, UV-Vis/Tauc-plot modeling, and dielectric/conductivity analysis across each doping series to build quantitative structure-property proxies for defect suppression.
- Engaged with DFT-based structural optimization, density-of-states, and band-structure outputs to interpret experimental trends (computational runs performed in collaboration; interpretive role).
- Presented research at SSCS Conference 2025 (2nd Prize, poster), ICRA-C 2024 (NUST Islamabad, oral), and ICC RTC&RET 2023 (University of Swat, poster); completed advanced analytical instrumentation training at the National Centre for Physics, Islamabad (2026).
- Mentored MS students on synthesis and characterization technique; served as a peer reviewer for the Journal of Materials Science: Materials in Electronics (Springer Nature).

MS Researcher, Chemistry

2017 – 2019

COMSATS University Islamabad, Abbottabad Campus

- Synthesized Mn/Ba-doped strontium zirconate (SrZrO_3) perovskites anchored on graphene oxide by coprecipitation and studied their visible-light photocatalytic performance against methylene blue dye degradation for wastewater treatment.
- Showed that substituting larger Ba^{2+} cations into SrZrO_3 enables bandgap engineering and antisite-defect formation, work that introduced defect-engineering and bandgap-tuning concepts later applied to CZTS.

Research Internship

Jul – Aug 2017

POF Institute of Technology, Pakistan Ordnance Factories, Wah Cantt (POF SJL Department)

PUBLICATIONS

FIRST AUTHOR · PHD RESEARCH

Investigation of the Optical Properties of Solution Processed Rare Metal-free Lithium Doped Kesterite Thin Films

Shahzad, W., Ismail, B.

Optical Materials, Vol. 170, 117582 (2026)

DOI: 10.1016/j.optmat.2025.117582

FIRST AUTHOR · PHD RESEARCH

Novel Insights from XRD and Photophysical Properties of $\text{Cu}_2\text{ZnSnS}_4$: Na Thin Films Deposited by Aqueous Chemical Route

Shahzad, W., Hassan, R., Badawi, A. K., Ismail, B.

Journal of Physics and Chemistry of Solids, Vol. 208, 113116 (2026)

DOI: 10.1016/j.jpcs.2025.113116

FIRST AUTHOR · MS RESEARCH

Enhanced Visible Light Photocatalytic Performance of $\text{Sr}_{0.3}(\text{Ba},\text{Mn})_{0.7}\text{ZrO}_3$ Perovskites Anchored on Graphene Oxide

Shahzad, W., Badawi, A. K., Rehan, Z. A., Khan, A. M., Khan, R. A., Shah, F., Ali, S., Ismail, B.

Ceramics International, Vol. 48, Issue 17, 24979–24988 (2022)

DOI: 10.1016/j.ceramint.2022.05.151

CO-AUTHOR

Transformation of Refractory Ceramic MgAl_2O_4 into Blue Light Emitting Nanomaterials by $\text{Sr}^{2+}/\text{Cr}^{3+}$ Activation

Chaudhary, A. R., Fazal, T., Jazaa, Y., Bahadur, A., Iqbal, S., Shah, M., Mahmood, S., Badawi, A. K., Khan, A. M., Shahzad, W., Ismail, B., Awwad, N. S., Ibrahim, H. A.

Materials Science and Engineering: B, Vol. 303, 117273 (2024)

DOI: 10.1016/j.mseb.2024.117273

CO-AUTHOR

Assessment of Thermodynamic Properties of SrSnO_3 Perovskite for Enhanced Photocatalytic Applications

Kanwal, S., Badawi, A. K., Khan, A. M., Khan, R. A., Shahzad, W., Ismail, B.

Journal of Inorganic and Organometallic Polymers and Materials, Vol. 35, 5767–5789 (2025)

DOI: 10.1007/s10904-025-03621-x

CONFERENCE PRESENTATIONS

SSCS Conference 2025 — Poster (2nd Prize)

April 2025

ICRA-C 2024, NUST Islamabad — Oral Presentation

2024

ICC RTC&RET 2023, University of Swat — Poster

2023

AWARDS & HONORS

- 2nd Prize, Poster Presentation — SSCS Conference, April 2025

PROFESSIONAL TRAINING & WORKSHOPS

Hands-on Training Workshop in Advanced Analytical Instrumentation

National Centre for Physics, Islamabad

Jun 29 – Jul 3, 2026

Three-Day Conference: Nanomaterials — New Trends in Development and Applications

Forman Christian College, Lahore

Jan 29–31, 2019

EMNT 2018 International Workshop on Energy Materials and Nanotechnology

COMSATS University Islamabad, Abbottabad Campus

Aug 15–16, 2018

PROFESSIONAL SERVICE

- Peer Reviewer, Journal of Materials Science: Materials in Electronics (Springer Nature)
- Mentored MS-level students in thin-film synthesis and characterization technique

RESEARCH TOOLKIT

Materials Synthesis: Chemical bath deposition · Two-step thermal annealing · Composition screening

Structural Analysis: XRD · Williamson-Hall analysis · Crystallite size, microstrain & dislocation density · VESTA modeling

Composition & Morphology: SEM · EDX/EDS · Grain size analysis

Optical Analysis: UV-Vis spectroscopy · Tauc analysis · Refractive index · Dielectric & conductivity modeling

Computational: DFT: structural optimization, DOS, band structure (interpretive engagement)

REFERENCES

Prof. Dr. Bushra Ismail

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bushraismail@cuiatd.edu.pk

Prof. Dr. Asad Muhammad Khan

Co-Supervisor, COMSATS University Islamabad, Abbottabad Campus

amk@cuiatd.edu.pk

Prof. Dr. Rafaqat Ali Khan

Supervisory Committee Member, COMSATS University Islamabad, Abbottabad Campus

rafaqatalikhan@cuiatd.edu.pk